

Lab experiment-7

Experiment 7: Breadth First Search (BFS)

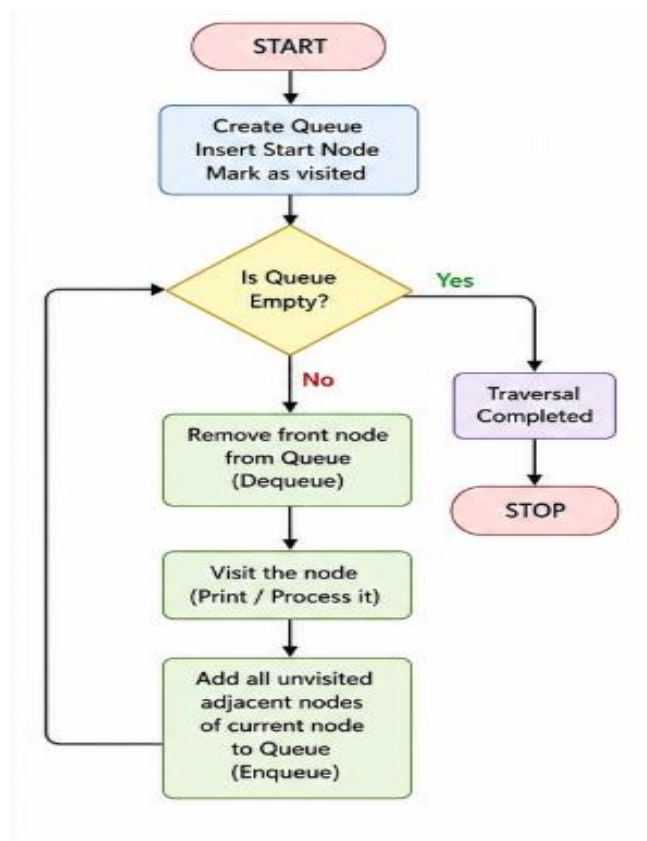
Aim

To implement Breadth First Search (BFS) using Python.

Objective

To traverse all nodes of a graph level by level using a queue.

Flow chart



Algorithm

1. Create a graph.
2. Insert the starting node into the queue.
3. Remove the front node.
4. Visit the node.
5. Add all unvisited neighboring nodes to the queue.
6. Repeat until the queue becomes empty.

Code

```
from collections import deque

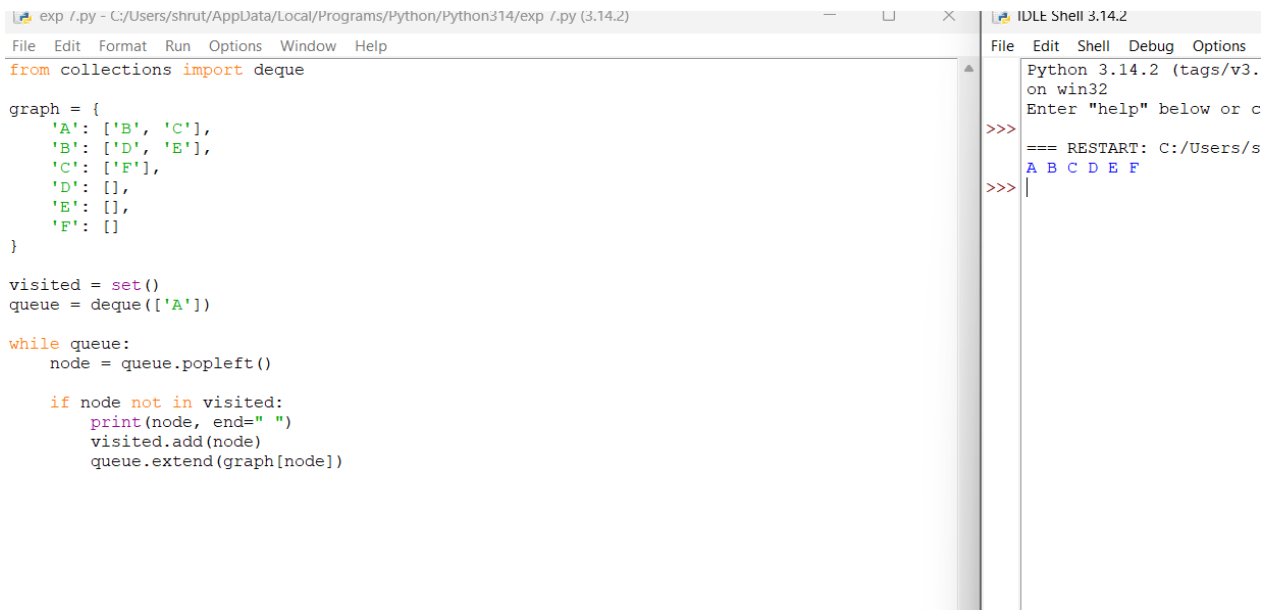
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []}

visited = set()

queue = deque(['A'])

while queue:
    node = queue.popleft()
    if node not in visited:
        print(node, end=" ")
        visited.add(node)
    queue.extend(graph[node])
```

Output



```
exp /_py - C:/Users/shrut/AppData/Local/Programs/Python/Python314/exp /_py (3.14.2)
File Edit Format Run Options Window Help
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

visited = set()
queue = deque(['A'])

while queue:
    node = queue.popleft()

    if node not in visited:
        print(node, end=" ")
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        queue.extend(graph[node])

IDLE Shell 3.14.2
File Edit Shell Debug Options
Python 3.14.2 (tags/v3.
on win32
Enter "help" below or c
>>>
=== RESTART: C:/Users/s
A B C D E F
>>>
```

Result

The graph was successfully traversed using Breadth First Search.

Conclusion

The BFS algorithm visits each node level by level using a queue and successfully traverses the graph.